

SNOWFLAKE
DATA CLOUD PLATFORM

USER MANUAL

Getting Started, Core Operations, and Administration Guide

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Document Control

Revision History

Version	Date	Author	Description of Change
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Document Approval

Role	Name	Date
Author	Mary Langley	2026-09-04
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Purpose and Scope

This manual provides end users, data analysts, data engineers, and administrators with the information needed to sign in to Snowflake, navigate the Snowsight web interface, create and manage core objects, load and query data, and apply role-based security. It is written for readers who are new to Snowflake as well as for existing users who need a consolidated reference.

This manual does not cover Snowflake's developer APIs, Snowpark programming, Snowflake Native App development, or cloud-provider-specific configuration in depth. Where relevant, it points to the areas of Snowflake's official documentation that cover those topics in more detail.

Intended Audience

- New Snowflake users signing in for the first time
- Data analysts and BI users who write and run SQL queries
- Data engineers who load, transform, and pipeline data
- Account and security administrators who manage users, roles, and warehouses

Responsibilities

Role	Responsibility
End User	Sign in, run queries, and manage personal worksheets within granted privileges.
Data Engineer	Create and maintain stages, file formats, pipelines, and transformation objects.
SYSADMIN	Create and manage warehouses, databases, and other account objects.
SECURITYADMIN	Manage grants, users, and roles; enforce the access control model.
ACCOUNTADMIN	Oversee billing, account-level configuration, and overall governance.

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1. Introduction to Snowflake

Snowflake is a cloud-based data platform, delivered as a fully self-managed service, that unifies data warehousing, data lake, data engineering, data science, and application workloads on a single system. Because Snowflake runs entirely on public cloud infrastructure, there is no hardware to size or install and effectively no software for account users to configure — Snowflake handles provisioning, patching, and tuning on the customer's behalf.

1.1 Platform Architecture

Snowflake's architecture blends the data-management simplicity of a shared-disk system with the scalability of a shared-nothing system. Queries run on massively parallel processing (MPP) compute clusters, while all persisted data lives in a central, cloud-hosted repository accessible to every compute cluster. The platform is organized into three logical layers:

1. Database storage — Snowflake automatically reorganizes loaded data into a compressed, columnar internal format, divided into units called micro-partitions. This layer supports standard Snowflake tables as well as Apache Iceberg tables (for data kept in externally managed cloud storage) and hybrid tables (optimized for high-throughput transactional access).
2. Compute — Virtual warehouses are independent clusters of compute resources that execute SQL statements and other workloads. Because each warehouse is isolated, one workload's activity does not degrade the performance of another.
3. Cloud services — A coordination layer that manages authentication, access control, metadata, query optimization, and infrastructure across the underlying cloud provider. This layer processes every request, from sign-in to query dispatch.

NOTE

Storage and compute are billed and scaled independently. You can grow your data volume without resizing compute, and vice versa.

1.2 Ways to Connect

Snowflake can be accessed through several interfaces, all of which reach the same underlying account and objects:

- **Snowsight** — the primary web-based interface, covered throughout this manual.
- **Snowflake CLI** — a command-line client for scripting and automation.
- **Native APIs** — the Snowflake Python API and Snowflake REST API for programmatic account and object management.
- **Drivers and connectors** — including JDBC, ODBC, and native connectors for tools such as Apache Kafka and Apache Spark.
- **Third-party tools** — ETL platforms and BI tools that connect to Snowflake through supported drivers.

2. Getting Started

2.1 Prerequisites

- A Snowflake account and an account identifier or account URL, provided by your administrator or from your trial confirmation email.
- A user name and either a password, single sign-on (SSO) credential, or key-pair/MFA method, depending on how your account is configured.
- A supported web browser for Snowsight, or a configured client (CLI, driver, or connector) if you plan to connect programmatically.

2.2 Signing In to Snowsight

Follow these steps to sign in for the first time:

1. Open a supported web browser and go to <https://app.snowflake.com>.
2. Enter your account identifier or account URL when prompted.
3. Enter your user name and authenticate using your configured method (password, SSO, or MFA).
4. On successful sign-in, Snowsight opens to the home page, showing your recently updated worksheets.

CAUTION

Running queries and most data-preview actions in Snowsight consume warehouse compute credits. Confirm which warehouse is active before running exploratory queries in a production account.

2.3 First-Time Orientation Checklist

- Confirm your default role and default warehouse with your administrator.
- Locate the databases and schemas you have been granted access to.
- Open a worksheet and run a simple test query, such as `SELECT CURRENT_ROLE(), CURRENT_WAREHOUSE();`
- Review the warehouse's auto-suspend setting so idle compute does not run unnecessarily.

3. Navigating the Snowsight Interface

Snowsight is Snowflake's web-based interface for data analysis, engineering, monitoring, and account administration. The interface is organized into four main areas:

Area	Purpose
Navigation menu	Create and manage worksheets, notebooks, databases, and other objects; access account administration.
Search	Locate databases, tables, views, functions, and other objects by name.
Quick actions	Perform common tasks for your current role, such as opening a worksheet or uploading a file to a table.
Recently viewed	Tabs showing recently accessed worksheets, notebooks, Streamlit apps, and dashboards.

3.1 Worksheets

Worksheets are the primary place to write and run SQL statements (and Python code, where enabled), review results, and visualize output. Each worksheet has a session context — an active role and an active warehouse — that determines what the worksheet can see and how its statements are billed.

- Open the Worksheets explorer to browse worksheets you own or that have been shared with you.
- Hover over a worksheet name to preview its contents and the role used to run it.
- Set or change a worksheet's role and warehouse from the session-context selector at the top of the worksheet.
- Use keyboard shortcuts (Ctrl+Shift+/ on Windows, Cmd+Shift+/ on Mac) to view the full shortcut list.

3.2 Notebooks and Other Work Surfaces

In addition to SQL worksheets, Snowsight provides Snowflake Notebooks for mixed SQL/Python analysis, Streamlit for building interactive data apps, and dashboards for combining multiple visualizations into a single view. These surfaces share the same underlying warehouses, roles, and data objects as worksheets.

4. Managing Databases, Schemas, and Tables

Snowflake organizes data in a strict container hierarchy: an account contains databases, each database contains one or more schemas, and each schema contains objects such as tables, views, stages, and file formats. Every object in this hierarchy is a securable object with a single owning role.

4.1 Creating a Database and Schema

```
CREATE DATABASE sales_db;  
CREATE SCHEMA sales_db.public;  
USE DATABASE sales_db;  
USE SCHEMA public;
```

4.2 Creating a Table

```
CREATE TABLE orders (  
  order_id      NUMBER,  
  customer_id   NUMBER,  
  order_date    DATE,  
  amount        NUMBER(10,2)  
);
```

When you load data into a Snowflake table, Snowflake automatically reorganizes it into compressed micro-partitions and manages file layout, compression, and statistics for you. You never need to define physical storage parameters.

4.3 Table Types

Table Type	Best Used For
Snowflake table (standard)	General-purpose structured and semi-structured data; the default choice for warehousing.
Apache Iceberg table	Data that must remain in externally managed cloud storage, such as an existing data lake.
Hybrid table	Transactional workloads that need row-level locking and low-latency single-row reads and writes.
Temporary / transient table	Short-lived data for staging or intermediate steps that does not need full Time Travel retention.

5. Virtual Warehouses

A virtual warehouse is a cluster of compute resources that supplies the CPU, memory, and temporary storage needed to run queries and data manipulation operations. Warehouses are either Standard or Snowpark-optimized (the latter recommended for memory-intensive workloads such as ML training). A warehouse must be running to execute a query, and it consumes Snowflake credits for every period it is active.

5.1 Creating and Sizing a Warehouse

```
CREATE WAREHOUSE analytics_wh
WAREHOUSE_SIZE = 'SMALL'
AUTO_SUSPEND = 60
AUTO_RESUME = TRUE;
```

Size	Typical Use
X-Small	Ad hoc queries, testing, and light development work.
Small – Medium	Standard analyst and BI workloads with moderate concurrency.
Large and above	Heavy transformation jobs, large joins, or high-concurrency workloads.

5.2 Managing Warehouse Activity

- `AUTO_SUSPEND` stops a warehouse after a period of inactivity so it stops consuming credits.
- `AUTO_RESUME` automatically restarts a suspended warehouse when a new query arrives.
- Multi-cluster warehouses add or remove clusters automatically to absorb concurrency spikes, such as during peak reporting hours.
- Each warehouse is isolated from every other warehouse, so resizing or heavily loading one warehouse does not affect the performance of others.

WARNING

Because compute time is billed per second (with a one-minute minimum) while a warehouse is running, leaving a large warehouse running unattended can generate unnecessary cost. Confirm `AUTO_SUSPEND` is set appropriately before leaving a session idle.

6. Loading Data into Snowflake

Snowflake refers to any location that holds data files awaiting load as a stage. Stages may be internal (managed inside your Snowflake account) or external (a cloud storage location you manage, such as Amazon S3, Google Cloud Storage, or Microsoft Azure Storage).

6.1 Stage Types

Stage Type	Description
User stage	Private storage allocated to a single user; files here can be loaded into multiple tables.
Table stage	Implicit storage tied to one table; not a separate object and has no independent grants.
Named internal stage	A schema-level database object that multiple users can stage files to and load from.
External stage	A schema-level object that points to a cloud storage location you manage outside Snowflake.

6.2 Loading Data with COPY INTO

The COPY INTO <table> command is the standard way to bulk-load data from a stage into a table. It supports common file formats, including CSV, JSON, Avro, ORC, Parquet, and XML.

```
CREATE STAGE my_stage
  URL = 's3://my-bucket/orders/'
  CREDENTIALS = (AWS_KEY_ID = '...' AWS_SECRET_KEY = '...');

COPY INTO orders
  FROM @my_stage
  FILE_FORMAT = (TYPE = CSV, SKIP_HEADER = 1);
```

6.3 Continuous and Streaming Loads

- Snowpipe loads new files automatically and with low latency as soon as they land in a stage — useful for near-real-time ingestion without manual COPY INTO runs.
- Snowpipe Streaming ingests row-level data directly into tables through an SDK or REST API, bypassing the file-staging step entirely.
- Openflow connectors and native Snowflake Connectors provide managed ingestion pipelines from specific external systems and applications.

6.4 Transforming Loaded Data

- Dynamic tables automatically refresh based on a defined query and a target freshness setting.
- Streams and tasks capture changes to a table and run scheduled transformation logic against those changes.
- Snowpark lets you write transformations in Python, Java, or Scala that run directly on warehouse compute.

7. Querying and Analyzing Data

7.1 Running a Query

Open a worksheet, confirm the active role and warehouse, and write a standard SQL statement. Results appear in a results pane below the worksheet, where you can filter, sort, and export them, or switch to a chart view to visualize the result set.

```
SELECT customer_id, SUM(amount) AS total_spend
FROM orders
GROUP BY customer_id
ORDER BY total_spend DESC
LIMIT 10;
```

7.2 Useful SQL Constructs

- Aggregate functions summarize a set of rows into a single value, such as SUM, AVG, or COUNT.
- Window functions perform calculations across a set of related rows without collapsing them into one row, useful for running totals or moving averages.
- Common table expressions (CTEs), written with the WITH clause, break a complex query into named, reusable steps for readability.

7.3 AI-Assisted Analysis

Snowflake Cortex provides AI functions that run large language models directly against data stored in Snowflake, enabling tasks such as summarization, translation, and free-form question answering over unstructured or semi-structured content without moving data to a separate platform. Snowflake ML provides tools for building and deploying your own models using the same governed data.

8. Security and Access Control

Snowflake's access control model combines role-based access control (RBAC), discretionary access control (each object has an owning role that can grant access), and, optionally, direct user-based grants. Access to any object is denied by default unless a grant explicitly allows it.

8.1 System-Defined Roles

Role	Purpose
ACCOUNTADMIN	Top-level role combining SYSADMIN and SECURITYADMIN; should be limited to a small number of trusted users.
SECURITYADMIN	Manages grants globally and creates, monitors, and manages users and roles.
USERADMIN	Dedicated to creating and managing users and roles.
SYSADMIN	Creates and manages warehouses, databases, and other account objects.
PUBLIC	Automatically granted to every user and role; objects it owns are visible to the whole account.

8.2 Custom Roles and Role Hierarchies

Administrators typically create custom roles scoped to specific business functions (for example, an ANALYST role or a DATA_ENGINEER role) and grant those roles up through a hierarchy that ultimately reports to SYSADMIN. This keeps account-management privileges separate from day-to-day object privileges, following Snowflake's recommended practice.

NOTE

A user can hold multiple roles but only one primary role is active at a time in a session (set with USE ROLE). Secondary roles can also be activated to combine privileges for a single session, such as cross-database joins.

8.3 Privileges

Privileges define the specific operations a role can perform on an object — for example, SELECT on a table, or USAGE on a schema or warehouse. Privileges are granted and revoked using GRANT and REVOKE statements, and future grants can be defined so that new objects created in a schema automatically inherit a standard set of privileges.

9. Sharing and Collaboration

Snowflake allows data to be shared across accounts without copying or moving it, which keeps a single governed copy of the data in sync everywhere it is consumed.

- Secure Data Sharing shares selected database objects directly with another Snowflake account.
- Listings let providers publish data and other assets, privately or through the Snowflake Marketplace, for other accounts to discover and use.
- Data Clean Rooms restrict shared data to a defined set of approved analyses, so a consumer can gain insight without gaining unrestricted access to the underlying rows.

Snowgrid, Snowflake's cross-region and cross-cloud technology layer, extends these capabilities across cloud providers and regions, applies consistent security and governance policies across that footprint, and supports replication for disaster recovery and business continuity.

10. Best Practices and Operational Notes

10.1 Cost Management

- Set `AUTO_SUSPEND` on every warehouse; an idle warehouse still consumes credits while running.
- Start with the smallest warehouse size that meets performance needs and scale up only when query queuing or slow execution demonstrates a real need.
- Use multi-cluster warehouses for concurrency spikes rather than permanently oversizing a single warehouse.

10.2 Data Governance

- Grant privileges to roles, not directly to users, and assign users to the appropriate roles — this keeps access auditable and easy to update.
- Reserve `ACCOUNTADMIN` for a small number of trusted administrators; use `SYSADMIN` and custom roles for day-to-day object management.
- Use future grants so that privileges are applied consistently to new objects as they are created.

10.3 Common Cautions

CAUTION

Dropping a database, schema, or table removes access to it immediately for all roles. Confirm you have the correct object and sufficient Time Travel retention configured before dropping production objects.

WARNING

Loading data with an incorrect `FILE_FORMAT` definition (wrong delimiter, encoding, or header setting) can silently load malformed rows. Validate a `COPY INTO` statement with the `VALIDATION_MODE` option before running it against a production table.

Appendix A: Glossary of Terms

Term	Definition
Account	The top-level container for all of an organization's databases, warehouses, users, and roles in Snowflake.
Micro-partition	A small, contiguous unit of compressed columnar storage that Snowflake automatically creates when data is loaded.
Virtual warehouse	A cluster of compute resources used to run queries and data manipulation operations.
Stage	A named location, internal or external, where data files are held before or after loading.
Role	An entity to which privileges on objects are granted; roles are assigned to users.
Privilege	A specific, grantable level of access to a securable object, such as SELECT or USAGE.
Snowsight	Snowflake's primary web-based user interface.
Worksheet	A Snowsight work surface for writing and running SQL or Python code.
Time Travel	A Snowflake feature that allows querying or restoring historical data within a retention window.
Snowpipe	A service that continuously loads new data files into a table as they arrive in a stage.

Appendix B: Snowsight Keyboard Shortcuts (Selected)

Action	Windows	Mac
Show all shortcuts	Ctrl+Shift+/ 	Cmd+Shift+/
Run current statement	Ctrl+Enter	Cmd+Enter
Format query	Shift+Ctrl+F	Shift+Cmd+F
Find and replace	Ctrl+H	Cmd+H
Comment / uncomment line	Ctrl+/ 	Cmd+/

Appendix C: First-Time Setup Checklist

- ☐ Confirm account identifier / account URL
- ☐ Verify sign-in method (password, SSO, or MFA)
- ☐ Confirm default role and default warehouse
- ☐ Confirm access to required databases and schemas
- ☐ Set AUTO_SUSPEND and AUTO_RESUME on any warehouse you create
- ☐ Run a test query to validate role and warehouse context
- ☐ Review the Security and Access Control section before requesting elevated privileges
- ☐ Bookmark <https://app.snowflake.com> and the Snowflake documentation portal for reference